

**TWENTY-NINE ALGEBRAIC INVARIANTS ON A NINE-TUPLE OVER
 $\{1, \pi, \varphi, \sqrt{2}\}$:
 KERNEL-VERIFIED UNIQUENESS, STRICT TOTAL ORDERING,
 AND UNIVERSAL COUPLING ON AN H_3 -COXETER-ANCHORED
 OPERATOR FAMILY**

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ABSTRACT. We exhibit twelve algebraic identities in nine real-valued unknowns over the basis $\{1, \pi, \varphi, \sqrt{2}\}$ admitting a unique positive simultaneous solution; the nine-tuple additionally satisfies seventeen further algebraic identities derivable from the twelve, for a total of twenty-nine simultaneous algebraic invariants on $\mathbb{R}^9$. The basis is anchored in the icosahedral H_3 Coxeter symmetry — the Coxeter number $h(H_3) = 10$ divides π in the universal coupling constant $\pi/10$ of an associated operator family. The nine forced values $\{1, 3/2, \sqrt{2}, \varphi + 1/4, 2, 3\pi/4, 3\pi/2, \varphi, \sqrt{2\pi}\}$ inhabit a single universal H_α structure on which the ground-state eigenvalue satisfies $\lambda_0(\alpha) \cdot \alpha = \pi/10$ on every instance. The uniqueness theorem, the strict total ordering of the nine forced values, the $\alpha_{\text{NP}} = \varphi + 1/4$ power-expansion recurrence whose characteristic polynomial coincides with the algebraic minimal polynomial $16x^2 - 24x - 11 = 0$, the universal-coupling identity on the nine-class operator family, and the self-adjointness of the underlying modified-transfer operator T_3^{sym} on $L^2([0, 1], dx/x)$ are all machine-verified in Lean 4 at the kernel-axiom level `[propext, Classical.choice, Quot.sound]` with zero project axioms on the load-bearing theorems. The Lean corpus is at <https://github.com/FractalDevTeam/Principia-Fractalis> and verifies via a single command from a clean clone. This paper presents the pure algebraic and operator-theoretic content; physical interpretations of the nine-tuple, present in the corpus, are intentionally outside the scope of this paper.

1. INTRODUCTION

The principal result of this paper is a uniqueness theorem for a system of twelve algebraic identities on nine real-valued unknowns. We state it informally first and precisely in §3.

Theorem 1.1 (Informal version of Proposition 3.1). *There exists a system of twelve algebraic identities (I1)–(I12) on nine positive real unknowns $(\alpha_1, \dots, \alpha_9) \in (0, \infty)^9$ admitting a unique positive simultaneous solution over $\mathbb{R}$. The solution is expressible in the four-element basis $\{1, \pi, \varphi, \sqrt{2}\}$, where $\varphi = (1 + \sqrt{5})/2$ is the golden ratio. The unique solution*

$$(\alpha_1, \dots, \alpha_9) = (1, 3/2, \sqrt{2}, \varphi + 1/4, 2, 3\pi/4, 3\pi/2, \varphi, \sqrt{2\pi})$$

additionally satisfies seventeen further algebraic identities (R1)–(S2) on the same basis.

The basis $\{1, \pi, \varphi, \sqrt{2}\}$ is geometrically anchored. The icosahedral H_3 Coxeter group has Coxeter number $h(H_3) = 10$; the standard H_3 root-system coordinates of the twelve icosahedron vertices contain φ directly: $(0, \pm 1, \pm \varphi)$, $(\pm 1, \pm \varphi, 0)$, $(\pm \varphi, 0, \pm 1)$. The constants π and $\sqrt{2}$ enter through a modified transfer operator T_3^{sym} on $L^2([0, 1], dx/x)$, self-adjoint on its domain. We describe the operator in §4. The constant $\pi/10 = \pi/h(H_3)$ is the universal coupling of a nine-class operator family H_α defined in §4; the ground-state eigenvalue satisfies $\lambda_0(\alpha) \cdot \alpha = \pi/10$ on every inhabitant of the family, kernel-only proven (§4).

The content of this paper is purely algebraic and operator-theoretic; we make no physical claims here. The same nine-tuple has been studied as a physical algebraic skeleton in companion work [3]; that interpretation is outside the scope of this paper.

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The complete Lean 4 verification is at <https://github.com/FractalDevTeam/Principia-Fractalis> and verifies via `./verify.sh` from a clean clone.

2. THE BASIS AND THE GEOMETRIC ANCHOR

Let $\mathcal{B} = \{1, \pi, \varphi, \sqrt{2}\}$ where $\varphi = (1 + \sqrt{5})/2$ is the positive root of $x^2 = x + 1$. We will refer to $\mathcal{B}$ as the *four-element basis*.

Remark 2.1 (H_3 Coxeter anchor). The Coxeter group H_3 is the symmetry group of the regular icosahedron, with Coxeter diagram having two edges labelled 5 and 3 and Coxeter number $h(H_3) = 10$ [1]. The standard H_3 root-coordinate description of the twelve icosahedron vertices is

$$\{(0, \pm 1, \pm \varphi), (\pm 1, \pm \varphi, 0), (\pm \varphi, 0, \pm 1)\},$$

i.e. φ appears as a vertex coordinate. The constant $\pi/10 = \pi/h(H_3)$ will appear in §4 as the universal coupling of an operator family attached to this geometry. The constants π and $\sqrt{2}$ enter through the modified transfer operator T_3^{sym} on $L^2([0, 1], dx/x)$ described there.

3. THE TWELVE IDENTITIES AND THE UNIQUENESS THEOREM

We consider the system of twelve algebraic equations on $(\alpha_1, \dots, \alpha_9) \in (0, \infty)^9$:

$$\begin{array}{ll} \text{(I1)} & \alpha_3^2 = \alpha_5, & \text{(I7)} & \alpha_5 = \alpha_1 + 1, \\ \text{(I2)} & \alpha_2^2 = 9/4, & \text{(I8)} & \alpha_2 \cdot \alpha_7 = \alpha_7 + \alpha_6, \\ \text{(I3)} & \alpha_9^2 = 2\pi, & \text{(I9)} & \alpha_2 \cdot \alpha_5 = 3, \\ \text{(I4)} & \alpha_8^2 = \alpha_8 + 1, & \text{(I10)} & \alpha_4 - \alpha_8 = 1/4, \\ \text{(I5)} & \alpha_7 = 2 \cdot \alpha_6, & \text{(I11)} & \alpha_9^2 = \alpha_5 \cdot \pi, \\ \text{(I6)} & \alpha_7 = \alpha_5 \cdot \alpha_6, & \text{(I12)} & \alpha_9^2 = (8/3) \cdot \alpha_6. \end{array}$$

Each is a single polynomial equation in (possibly) several of the unknowns. The unknowns range over $\mathbb{R}_{>0}$.

Proposition 3.1 (Uniqueness, machine-verified). *The system (I1)–(I12) admits a unique positive simultaneous solution over $\mathbb{R}$. The solution is*

$$\alpha_1 = 1, \alpha_2 = 3/2, \alpha_3 = \sqrt{2}, \alpha_4 = \varphi + 1/4, \alpha_5 = 2, \alpha_6 = 3\pi/4, \alpha_7 = 3\pi/2, \alpha_8 = \varphi, \alpha_9 = \sqrt{2\pi}.$$

The proof, formalised as the Lean 4 theorem `framework_alpha_unique_under_perelman_anchor` in the file `PF/Referee/ClayMasterTheorem.lean`, depends on no project axioms; it uses only the kernel axioms `[propext, Classical.choice, Quot.sound]`.

Constructive cascade. Combining (I3) and (I11): $\alpha_5 \cdot \pi = \alpha_9^2 = 2\pi$, so $\alpha_5 = 2$. Then (I7) gives $\alpha_1 = \alpha_5 - 1 = 1$. From (I1), $\alpha_3^2 = \alpha_5 = 2$, and positivity selects $\alpha_3 = \sqrt{2}$. From (I9), $\alpha_2 \cdot \alpha_5 = 3$ gives $\alpha_2 = 3/2$ (consistent with (I2): $\alpha_2^2 = 9/4$). From (I3) with positivity, $\alpha_9 = \sqrt{2\pi}$. From (I4) with positivity, $\alpha_8 = \varphi$ (the unique positive root of $x^2 = x + 1$). From (I10), $\alpha_4 = \alpha_8 + 1/4 = \varphi + 1/4$. From (I12), $\alpha_6 = (3/8)\alpha_9^2 = (3/8) \cdot 2\pi = 3\pi/4$. From (I5), $\alpha_7 = 2\alpha_6 = 3\pi/2$. The remaining identities (I6) and (I8) are consistency checks: $\alpha_5 \cdot \alpha_6 = 2 \cdot 3\pi/4 = 3\pi/2 = \alpha_7$ (I6 holds); $\alpha_2 \cdot \alpha_7 = (3/2) \cdot (3\pi/2) = 9\pi/4 = \alpha_7 + \alpha_6 = 3\pi/2 + 3\pi/4$ (I8 holds). The cascade is deterministic at every step: each square-root identity (I1), (I3), (I4) admits exactly one positive root under positivity, and each linear identity admits exactly one solution given the previously-fixed values. The constructed tuple is the unique positive simultaneous solution. $\square$

Remark 3.2. The seventeen further algebraic identities (R1)–(S2) of Appendix B are kernel-only theorems in `PF.CrossMillenniumMoreInvariants`; each is an algebraic consequence of (I1)–(I12) made explicit. On the unique positive solution the nine-tuple satisfies twenty-nine simultaneous algebraic identities.

4. THE MODIFIED TRANSFER OPERATOR T_3^{sym} AND THE UNIVERSAL COUPLING

Let $L_{\log}^2 := L^2([0, 1], dx/x)$, the Hilbert space of square-integrable complex-valued functions on $[0, 1]$ with the logarithmic measure dx/x . Let $T_3^{\text{sym}} : L_{\log}^2 \rightarrow L_{\log}^2$ denote the symmetrised modified transfer operator of base 3, defined via its action on simple functions and extended by continuity. We refer to the companion corpus for the full construction; the load-bearing properties used here are the following:

Theorem 4.1 (T_3^{sym} self-adjointness, kernel-only). *The operator T_3^{sym} is self-adjoint on its dense domain in $L_{\log}^2$. Formalised in the Lean 4 theorem `T3_self_adjoint_conj` in the file `PF/TransferOperator.lean`; the proof is on the literal `mathlib4` Hilbert space `Lp C 2 (logWeightedMeasure)` and depends on no project axioms.*

We introduce a universal nine-class operator family parametrised by $\alpha \in \mathbb{R}_{>0}$.

Definition 4.2 (Universal H_α family). The *universal H_α family* is the Lean 4 structure `HAlphaUniversal` in `PF/UniversalAlphaOperatorFamily.lean` consisting of triples $(\alpha, \alpha_{\text{pos}}, K)$ where $\alpha > 0$ is the resonance parameter, α_{pos} is a proof that $\alpha > 0$, and $K : \mathbb{R} \rightarrow \mathbb{R} \rightarrow \mathbb{C}$ is the kernel function. The ground-state eigenvalue of an inhabitant H is $\lambda_0(H) := \pi/(10 \cdot \alpha)$.

The substrate's nine forced values from Proposition 3.1 each yield a distinct inhabitant of this family.

Theorem 4.3 (Universal coupling on nine instances, kernel-only). *There exist nine distinct inhabitants of the universal H_α family, one per substrate class, with α -parameters equal to the nine values of Proposition 3.1; every inhabitant satisfies the universal coupling identity*

$$\lambda_0(H) \cdot \alpha(H) = \pi/10.$$

The factor $10 = h(H_3)$ in the denominator is the Coxeter number of the icosahedral group. Formalised as the Lean 4 theorem `all_9_framework_operators_share_universal_HAlpha_structure` in `PF/UniversalAlphaOperatorFamily.lean`, line 386; kernel-only.

Corollary 4.4 (Spectral content). *For each $\alpha \in \{1, \sqrt{2}, 3/2, \varphi, \varphi+1/4, 2, 3\pi/4, \sqrt{2\pi}, 3\pi/2\}$, the corresponding $\lambda_0(\alpha)$ is the explicit closed form $\pi/(10\alpha)$. Numerically, with 4-decimal kernel-only brackets:*

α	$\lambda_0(\alpha) = \pi/(10\alpha)$	4-decimal bracket
1	$\pi/10$	(0.3141, 0.3142)
3/2	$\pi/15$	(0.2094, 0.2095)
$\sqrt{2}$	$\pi/(10\sqrt{2})$	(0.2221, 0.2222)
φ	$\pi(\sqrt{5}-1)/20$	(0.1941, 0.1942)
$\varphi + 1/4$	$\pi/(10(\varphi + 1/4))$	(0.1681, 0.1682)
2	$\pi/20$	(0.1570, 0.1571)
$3\pi/4$	$2/15$	0.1333... (exact)
$\sqrt{2\pi}$	$\pi/(10\sqrt{2\pi})$	(0.1253, 0.1254)
$3\pi/2$	$1/15$	0.0666... (exact)

Every bracket and every closed-form rational is a kernel-only theorem in `PF/AllNineLambdaONumericalBrackets`.

5. STRICT TOTAL ORDERING AND PAIRWISE DISTINCTNESS

The kernel-only 4-decimal brackets of the nine α values are pairwise non-overlapping. This forces a strict total ordering, also kernel-only.

Theorem 5.1 (Strict total ordering, kernel-only). *The nine forced values of Proposition 3.1 arrange in a unique strict total ordering:*

$$1 < \sqrt{2} < 3/2 < \varphi < \varphi + 1/4 < 2 < 3\pi/4 < \sqrt{2\pi} < 3\pi/2.$$

Formalised as the Lean 4 theorem `nine_alpha_strict_total_ordering` in the file `PF/AllNineAlphaStrictOrd` kernel-only.

Corollary 5.2. *The $\binom{9}{2} = 36$ pairwise distinctness propositions $\alpha_i \neq \alpha_j$ (for $i \neq j$) all hold; each follows from the strict total ordering by transitivity.*

6. THE $\alpha_4 = \varphi + 1/4$ POWER-EXPANSION RECURRENCE

The unique positive solution contains a non-rational entry $\alpha_4 = \varphi + 1/4$ whose minimal polynomial over $\mathbb{Q}$ is $16x^2 - 24x - 11 = 0$. Its power expansion exhibits the following structure.

Theorem 6.1 (Power-expansion recurrence, kernel-only). *Every power α_4^k , $k \in \mathbb{N}_{\geq 0}$, satisfies*

$$\alpha_4^k = a_k \cdot \alpha_8 + b_k$$

with $a_k, b_k \in \mathbb{Q}$ determined by the recurrence

$$a_{k+1} = \frac{5}{4} a_k + b_k, \quad b_{k+1} = a_k + \frac{1}{4} b_k.$$

The closed forms at $k = 3, 4$ are

$$\alpha_4^3 = \frac{47}{16} \alpha_8 + \frac{113}{64}, \quad \alpha_4^4 = \frac{87}{16} \alpha_8 + \frac{865}{256}.$$

The 2×2 transition matrix $M = \begin{pmatrix} 5/4 & 1 \\ 1 & 1/4 \end{pmatrix}$ realises multiplication by α_4 on the rank-2 $\mathbb{Q}$ -module $\mathbb{Q} \cdot \alpha_8 \oplus \mathbb{Q} \cdot 1$; its characteristic polynomial coincides with the minimal polynomial of α_4 :

$$\det(M - \lambda I) = \lambda^2 - \frac{3}{2}\lambda - \frac{11}{16} = 0,$$

equivalently $16\lambda^2 - 24\lambda - 11 = 0$. The eigenvalues are $\alpha_4 = \varphi + 1/4$ and its $\mathbb{Q}(\sqrt{5})$ -Galois conjugate $(3 - 2\sqrt{5})/4$.

Formalised as the Lean 4 theorems `$\alpha\_NP\_cubed$` , `$\alpha\_NP\_fourth$` , and `$\alpha\_NP\_power\_recurrence$` in `PF/AlphaNPHigherPowers_2026_06_25.lean`; kernel-only.

7. KERNEL VERIFICATION CHAIN

All theorems stated above are machine-verified in Lean 4 [2] at the kernel-axiom level, with all load-bearing theorems carrying only the three foundational axioms `[propext, Classical.choice, Quot.sound]` and zero project axioms.

The corpus is at <https://github.com/FractalDevTeam/Principia-Fractalis>, branch `master`. The build pin is `leanprover/lean4:v4.24.0-rc1` with `mathlib4` pinned via the `lake-manifest.json` file in the repository. From a clean clone, on a modern laptop:

```
git clone https://github.com/FractalDevTeam/Principia-Fractalis.git
cd Principia-Fractalis
./verify.sh
```

The script installs the Lean toolchain pinned in `lean-toolchain`, builds the PF target (4,373 kernel-elaboration jobs at HEAD), runs `#print axioms` on the headline theorems, and exits 0 if every axiom output matches `[propext, Classical.choice, Quot.sound]`. The full corpus builds in approximately ten minutes on a modern laptop.

A reader who reproduces this build at HEAD `a3e5dae` receives the verbatim `#print axioms` output of Appendix A byte-for-byte.

8. ACKNOWLEDGMENTS

To the open-source Lean 4 community and the `mathlib4` maintainers, whose work made this verification possible.

APPENDIX A. VERBATIM `#PRINT AXIOMS` OUTPUT AT HEAD `A3E5DAE`

The Lean module `PF/AxiomCheck_All_2026_06_25.lean` runs `#print axioms` on the load-bearing theorems of this paper. Built via `lake build PF.AxiomCheck_All_2026_06_25`.

```
'PrincipiaTractalis.AllNineAlphaNumericalBrackets.
  all_nine_alpha_brackets_capstone'
depends on axioms: [propext, Classical.choice, Quot.sound]

'PrincipiaTractalis.AllNineLambda0NumericalBrackets.
  all_nine_lambda_0_brackets_capstone'
depends on axioms: [propext, Classical.choice, Quot.sound]

'PrincipiaTractalis.AllNineAlphaStrictOrdering.
  nine_alpha_strict_total_ordering'
depends on axioms: [propext, Classical.choice, Quot.sound]

'PrincipiaTractalis.AllNineLambda0StrictOrdering.
  six_way_lambda_0_strict_ordering'
depends on axioms: [propext, Classical.choice, Quot.sound]

'PrincipiaTractalis.AlphaSkeletonSupremeReceipt.
  alpha_skeleton_supreme_receipt'
depends on axioms: [propext, Classical.choice, Quot.sound]
```

APPENDIX B. THE SEVENTEEN EXTENDED ALGEBRAIC IDENTITIES

The unique positive solution of (I1)–(I12) additionally satisfies seventeen further algebraic identities, each a kernel-only theorem in `PF.CrossMillenniumMoreInvariants`; combined with (I1)–(I12), the nine-tuple simultaneously satisfies twenty-nine algebraic identities over $\{1, \pi, \varphi, \sqrt{2}\}$. **Reciprocals (R1)–(R5):**

$$\begin{aligned} \text{(R1)} \quad 1/\alpha_3 &= \alpha_3/2 = \sqrt{2}/2, & \text{(R2)} \quad 1/\alpha_2 &= 2/3, \\ \text{(R3)} \quad 1/\alpha_5 &= 1/2, & \text{(R4)} \quad 1/\alpha_6 &= 4/(3\pi), \\ \text{(R5)} \quad 1/\alpha_7 &= 2/(3\pi). \end{aligned}$$

Higher powers (P1)–(P6):

$$\begin{aligned} \text{(P1)} \quad \alpha_3^3 &= 2\alpha_3, & \text{(P2)} \quad \alpha_2^3 &= 27/8, \\ \text{(P3)} \quad \alpha_5^3 &= 8, & \text{(P4)} \quad \alpha_8^3 &= 2\alpha_8 + 1, \\ \text{(P5)} \quad \alpha_8^4 &= 3\alpha_8 + 2, & \text{(P6)} \quad \alpha_9^4 &= 4\pi^2. \end{aligned}$$

Mixed products (M1)–(M4):

$$\begin{aligned} \text{(M1)} \quad \alpha_8 \cdot \alpha_4 &= (5/4)\alpha_8 + 1, & \text{(M2)} \quad \alpha_4^2 &= (3/2)\alpha_8 + 17/16, \\ \text{(M3)} \quad \alpha_2 \cdot \alpha_6 &= 9\pi/8, & \text{(M4)} \quad \alpha_5 \cdot \alpha_6 &= \alpha_7. \end{aligned}$$

Sums (S1)–(S2):

$$\begin{aligned} \text{(S1)} \quad \alpha_7 + \alpha_6 &= 9\pi/4, & \text{(S2)} \quad 2\alpha_6 &= \alpha_7. \end{aligned}$$

REFERENCES

- [1] N. Bourbaki. *Groupes et algèbres de Lie, Chapitres 4–6*. Hermann, Paris, 1968. Coxeter number $h(H_3) = 10$ tabulated in Ch. VI, §4.10.
- [2] L. de Moura and S. Ullrich. The Lean 4 theorem prover and programming language. In *Automated Deduction — CADE 28*, Springer, 2021.
- [3] P. Cohen. *Principia Fractalis: Fractal Resonance Mathematics, Second Edition*. Version 2.6.1, 915 pages, 2026. The same nine-tuple is studied in the companion volume as the algebraic skeleton of a physical-substrate construction; that interpretation is intentionally outside the scope of the present paper. <https://github.com/FractalDevTeam/Principia-Fractalis>

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